67a3c6ac	Transformation applied: 1.Rotate the colors clockwise within the objects.	✗
a5313dff	Transformation applied: 1.Recolor color 0 objects to color 1	✓
a699fb00	Transformation applied: 1. Recolor color 0 objects to color 2	✓

E Logistic regression analysis

Feature	**Coefficient**	
	Direct-grid	Object-based
Number of colored pixels in test input image	-0.151312	-0.365261
Average number of colored pixels in training input images	0.215891	0.326572
Number of unique colors in test input image	-0.282226	0.346230
Average number of unique colors in training input images	0.192485	-1.186780
Number of pixels changed in test instance	0.110529	0.142800
Average number of pixels changed in training instances	-0.152656	-0.090327
Test input image size	-0.004665	0.001771
Training input images average size	-0.013070	-0.005959
Number of training instances	0.297392	0.158643
Performance Metric	**Score**	
	Direct-grid	Object-based
Precision (unsolved)	0.78	0.83
Precision (solved)	0.44	0.73
Recall (unsolved)	0.86	0.74
Recall (solved)	0.31	0.83

Table 8: **Results of logistic regression analysis.** Top: Comparison of feature coefficients for the best-performing direct-grid and object-based approaches, demonstrating the impact of each feature on an ARC task's solvability. Bottom: Precision and recall scores of logistic regression model for solved and unsolved tasks.

F Visualization of GPT solutions on example ARC tasks

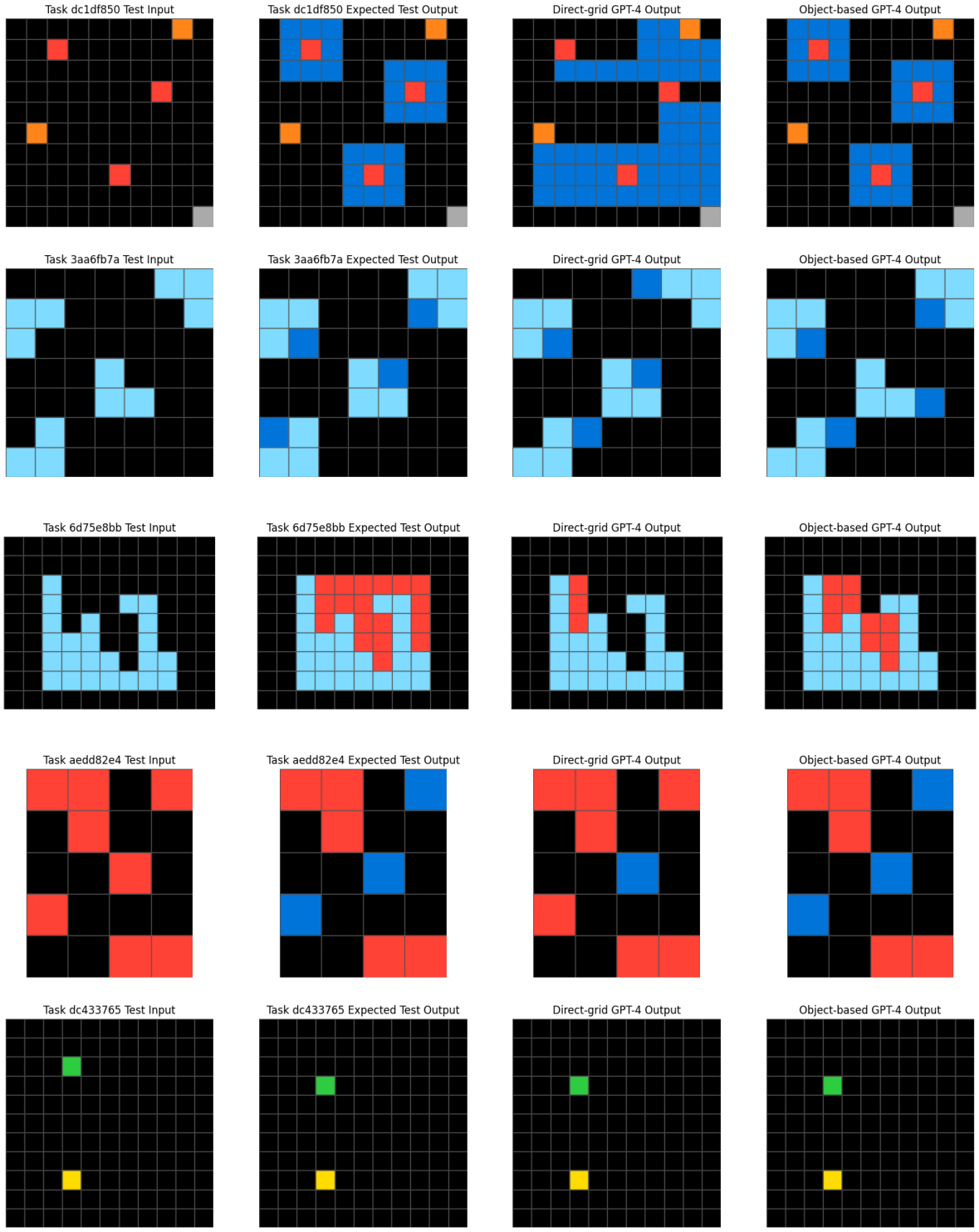

Figure 13: **Visualization of example ARC Tasks.** Each row showcases an ARC task. The first column displays the test input for that task, while the second column shows the expected test output. The third and fourth columns present the predicted outputs using the best-performing direct-grid approach and object-based approach, respectively.

G Experiments with GPT-4V

This section briefly introduces the preliminary experiments conducted for GPT-4V. The GPT-4V API was not available at the time of our research, therefore, the experiments were conducted by manually prompting ChatGPT-4 with image attachments. First, the prompting method is described and visualized, followed by a display of the results. Examples of GPT's output will also be shown.

G.1 Prompting

The prompting style follows the few-shot approach introduced in Section 2.2. It consists of "instructions" and "task" sections. The "instructions" section is similar to before where it outlines the nature of an ARC task and the expected behavior of the LLM. The "task" section provides information about the ARC task of interest, however, instead of using textual representations, an image containing the few-shot examples as well as the test input is provided. Figure 14 shows an example of the prompts.

G.2 Results

GPT-4V cannot currently produce images as outputs, so it produces texts that describe the output grid. To evaluate the results, we manually examined GPT-V4's outputs and reconstructed the output images based on the descriptions it produced. We found that GPT-4V was only able to correctly describe the output image for 2 tasks out of the 50 tasks we defined earlier, even more so, only a further 3 out of 50 tasks had the correct logic before producing the wrong output. 2 examples of ARC tasks as well as GPT-4V outputs for them can be seen in Figure 15